

A note on electric field

→ Electric field produced by charges

ELECTROSTATICS

$$\nabla \times \mathbf{E} = 0 \quad \forall \mathbf{r} \in D,$$

D is SCD



→ Conservative field

→ Can be written as $\mathbf{E} = -\nabla V$

→ Zero circulation around any closed path

$$\oint_{\gamma} \mathbf{E} \cdot \hat{\mathbf{l}} \, dl = 0$$

→ Electric field produced by electromagnetic induction

ELECTRODYNAMICS

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

→ Non-conservative field

→ Non-zero circulation may exist around closed paths

$$\oint_{\gamma} \mathbf{E} \cdot \hat{\mathbf{l}} \, dl = \varepsilon$$

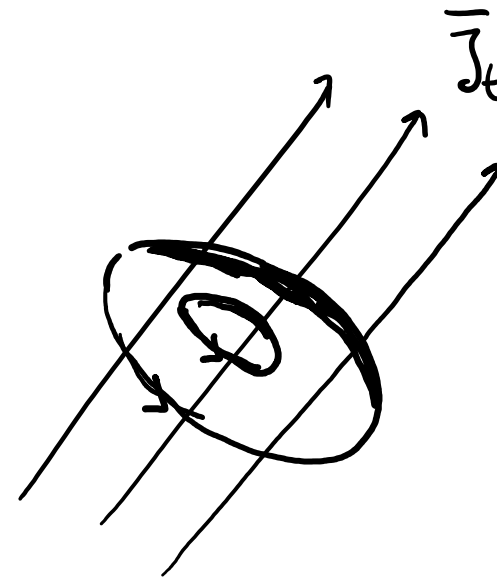
Charges produce the **conservative part** of the electric field, while time-varying magnetic fields produce a **non-conservative circulating component**

Ampere-Maxwell law – local form

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) = \underbrace{\mathbf{J}}_{\substack{\text{CONDUCTION} \\ \text{CURRENT DENSITY} \\ [\text{A/m}^2]}} + \underbrace{\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}}_{\substack{\text{DISPLACEMENT} \\ \text{CURRENT DENSITY} \\ [\text{A/m}^2]}}$$

$\rightarrow$ The total current density produces a circulating magnetic field

$\underbrace{\hspace{10em}}_{\substack{\mathbf{J}_t \\ \text{Total current density}}}$



Ampere-Maxwell law – integral form

Consider open surface S bounded by curve Γ_S

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) = \mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

integrate both sides over S

$$\int_S \nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) \cdot \hat{\mathbf{n}} dS = \int_S \mathbf{J} \cdot \hat{\mathbf{n}} dS + \epsilon_0 \int_S \frac{\partial \mathbf{E}}{\partial t} \cdot \hat{\mathbf{n}} dS$$

apply Stokes theorem

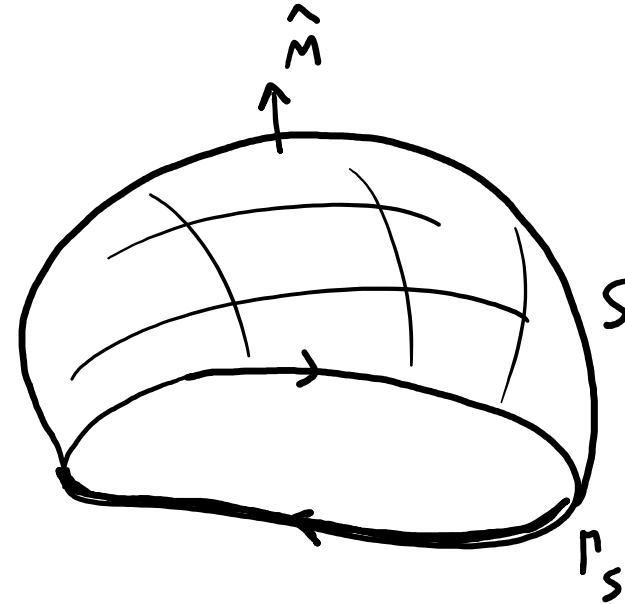
$$\oint_{\Gamma_S} \frac{\mathbf{B}}{\mu_0} \cdot d\mathbf{l} =$$

MAGNETOMOTIVE
FORCE
(mmf)

$$i_c + i_D$$

$$\boxed{\mathcal{F} = i_T}$$

S is STILL
 $\epsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot \hat{\mathbf{n}} dS$



$\mathcal{F}$ [A]: magnetomotive force

circulation of magnetic field intensity
along a closed path

Faraday-Neumann-Lenz law – integral form

$$\boxed{\mathcal{F} = i_T}$$

An electric current through a surface produces a circulation of the magnetic field intensity around its boundary

Exercise: show that $\mathbf{J}_T$ is a solenoidal vector field

$$\underbrace{\nabla \cdot [\nabla \times (1/\mu_0 \bar{\mathbf{B}})]}_{\nabla \cdot (\nabla \times (\dots)) = 0} = \underbrace{\nabla \cdot [\bar{\mathbf{J}} + \epsilon_0 \frac{\partial \bar{\mathbf{E}}}{\partial t}]}_{\bar{\mathbf{J}}_T} \Rightarrow \boxed{\nabla \cdot \bar{\mathbf{J}}_T = 0} \quad \forall \bar{\mathbf{r}} \in D$$

↓ if D is SCD

$$\boxed{\oint_S \bar{\mathbf{J}}_T \cdot \hat{\mathbf{n}} dS = 0}$$

$\bar{\mathbf{J}}_T$ is a SOLENOIDAL FIELD

Current continuity equation

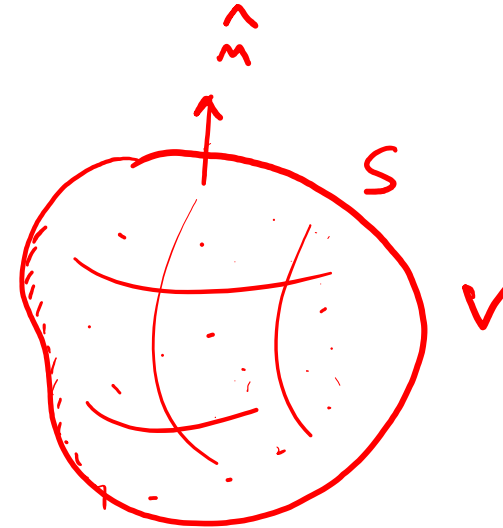
$$\text{LOCAL: } \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0 \Rightarrow \frac{\partial \rho}{\partial t} = -\nabla \cdot \vec{J}$$

↑
CHANGES in time of charge density

$$\int_V \frac{\partial \rho}{\partial t} dV = - \int_V \nabla \cdot \vec{J} dV$$

$$\frac{d}{dt} \underbrace{\int_V \rho dV}_{Q_{ENC}} = - \underbrace{\oint_S \vec{J} \cdot \hat{n} dS}_{\dot{\lambda}} \Rightarrow$$

$$\boxed{\frac{dQ_{ENC}}{dt} = -\dot{\lambda}}$$



Electromagnetics in matter

Electric polarization (context)

Focus: electric fields in matter

→ **Conductors**

electrons not associated with specific nucleus → **free** to move

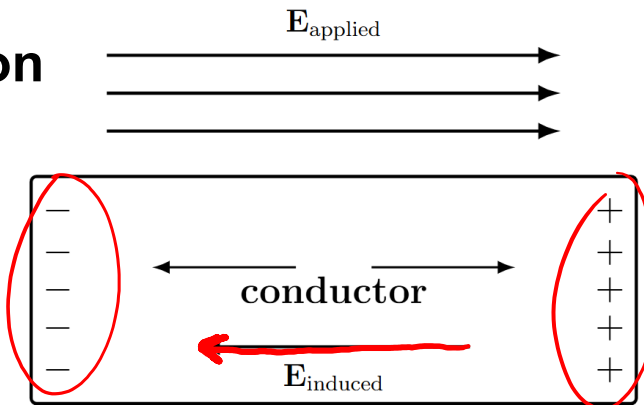
→ **Insulators (dielectrics)**

charges **bound** to specific atoms or molecules

response to applied electric field

macroscopic **redistribution of free charge**

screening of internal field (at equilibrium)



microscopic **displacements** of bound charges within material

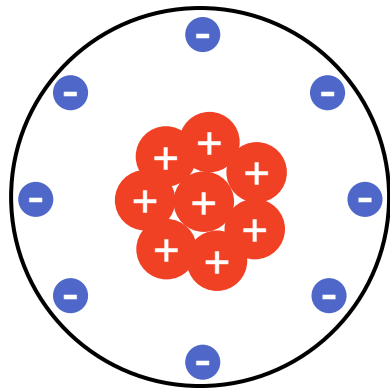
electric polarization

Electric polarization

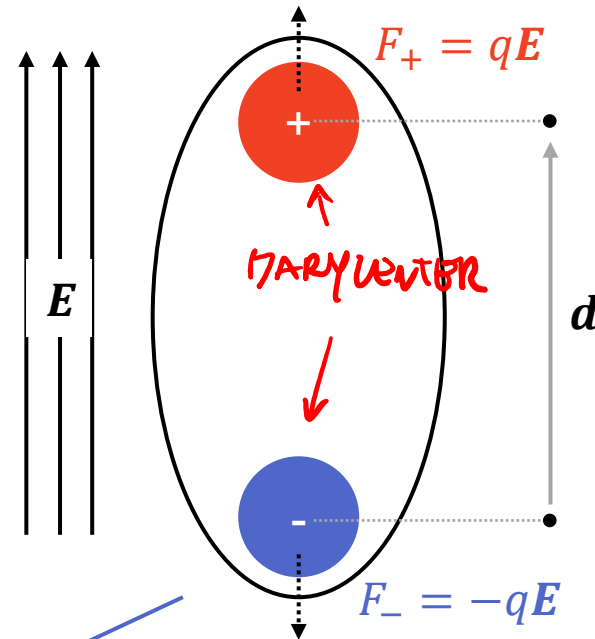
Mechanism 1 – **induced** polarization (non-polar matter)

e.g., noble gases, N_2 , O_2 , hydrocarbons...

no external field



with external field



elementary electric dipole

electric dipole moment

A diagram showing the electric dipole moment vector $\mathbf{p}$. It is represented by a vertical black arrow pointing upwards and a red arrow pointing downwards. The equations $\mathbf{p} = q\mathbf{d}$ and $[p] = \text{C m}$ are written next to the arrows.

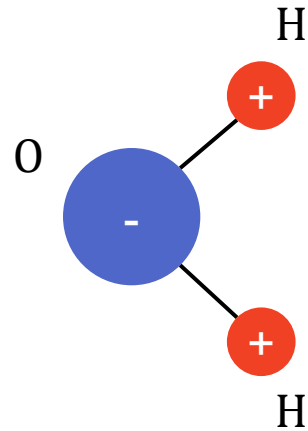
Recall: for $r \gg d$,

$$\varphi(\mathbf{r}) \simeq \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$$

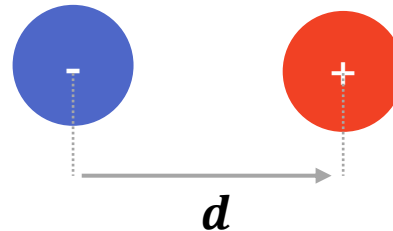
Electric polarization

Mechanism 2 – polarization by orientation (polar matter) e.g., H_2O , HCl , NH_3 , (many) polymers...

no external field



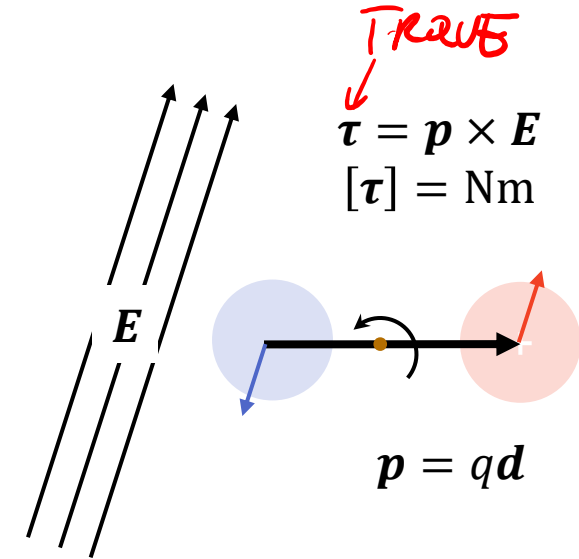
DIPOLES



*elementary electric dipole
(in absence of external fields)*



with external field



*external field exerts a
torque τ that aligns the
dipole*

Electric polarization field

Goal: obtain **macroscopic description** of polarization

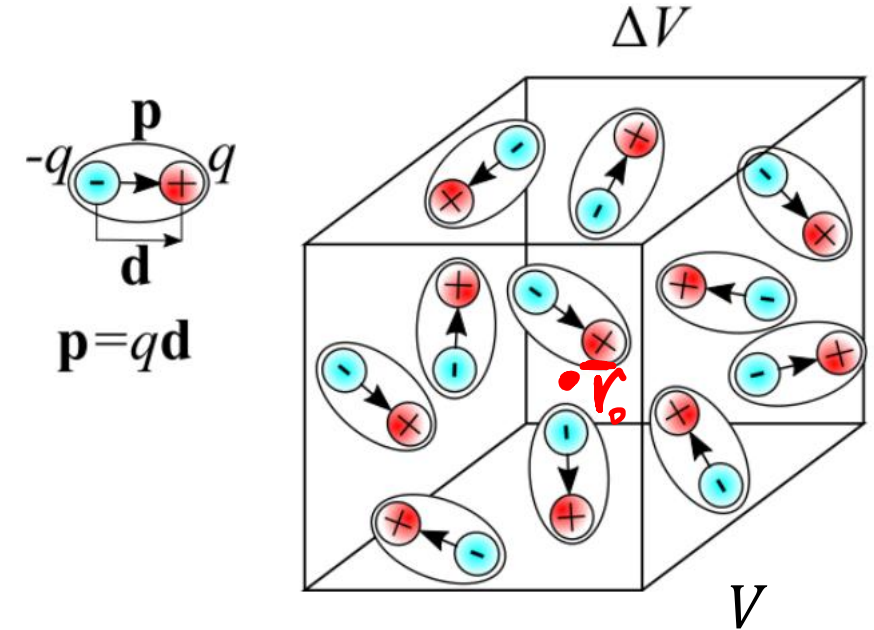
Let $\mathbf{r}_0$ be a point in a region V with volume ΔV

$$\mathbf{P}(\mathbf{r}_0) = \lim_{\Delta V \rightarrow 0} \frac{\text{macro } \Delta \mathbf{p}}{\Delta V} \quad [\mathbf{P}] = \frac{\text{C m}}{\text{m}^3} = \frac{\text{C}}{\text{m}^2}$$

(Note: In the original image, the word 'macro' and the units 'C/m²' are circled in red, and the equation $\Delta \mathbf{p} = \sum_{i \in \Delta V} \mathbf{p}_i$ is also circled in red.)

Polarization field: electric dipole moment per unit volume
measures the local strength and direction of electric polarization

Problem: polarization redistributes bound charge inside matter, and this charge contributes to $\mathbf{E}$



Electric polarization field

A distribution of electric dipoles can be represented by an *equivalent* volume charge

$$\boxed{-\nabla \cdot \mathbf{P} = \rho_p}$$

Handwritten notes: A red arrow points to the ρ_p term with the label $4\pi^2$. The entire equation is enclosed in a red circle.

ρ_p : polarization charge density [C/m³]

...we can fix the Gauss' law for electric fields!

in free space

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\epsilon_0}$$

Handwritten notes: The equation is circled in red. Above it, a red line points to the text $\rho = \rho_f$.

in free space all
charge is free charge

in (polarizable) matter

$$\nabla \cdot \mathbf{E} = \frac{\rho_f + \rho_p}{\epsilon_0}$$

Handwritten notes: The equation is circled in red.

$$\rho = \rho_f + \rho_p$$

Handwritten notes: The equation is circled in red. Two blue lines connect this equation to the two Gauss's law equations above it.

Electric polarization field

$$\rightarrow \nabla \cdot \mathbf{E} = \frac{\rho_f + \rho_p}{\epsilon_0}$$

$$-\nabla \cdot \mathbf{P} = \rho_p$$

$$\nabla \cdot (\epsilon_0 \mathbf{E}) = \rho_f - \nabla \cdot \mathbf{P}$$

$$\rightarrow \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f$$

$\mathbf{D}$ Electric displacement field [C/m²]

$$\nabla \cdot (\epsilon_0 \mathbf{E}) = \rho_f \quad \text{in free space}$$

$$\nabla \cdot \mathbf{D} = \rho_f \quad \text{in matter}$$

in free space

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \cancel{\mathbf{P}} = \epsilon_0 \mathbf{E}$$

$\mathbf{D}$ is identical to $\mathbf{E}$, just scaled by ϵ_0

in matter

$$\rightarrow \mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

$\mathbf{D}$ is a function of $\mathbf{E}$: $\mathbf{P} = \mathbf{P}(\mathbf{E}) \rightarrow \mathbf{D} = \mathbf{D}(\mathbf{E})$

Electric polarization field – linear and isotropic materials

General case:

Pol. along x induced by E_x

$$\rightarrow \boxed{\mathbf{P} = \varepsilon_0 \boldsymbol{\chi} \mathbf{E}} \rightarrow \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \varepsilon_0 \begin{bmatrix} \chi_{xx} & \chi_{xy} & \chi_{xz} \\ \chi_{yx} & \chi_{yy} & \chi_{yz} \\ \chi_{zx} & \chi_{zy} & \chi_{zz} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$

χ : electric polarization tensor

polarization induced along z by E_x

polarization induced along z by E_z

$$\nabla \cdot (\varepsilon_0 \bar{\mathbf{E}} + \bar{\mathbf{P}}) = \int f$$

$\uparrow$
 $\varepsilon_0 \bar{\boldsymbol{\chi}} \bar{\mathbf{E}}$

Electric polarization field – linear and isotropic materials

For linear, isotropic materials

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$$

→ χ_e : electric susceptibility [–]
characteristic of specific material

ε_r	free-space	= 1
	air	~ 1
	solid dielectrics	~ 2 ÷ 6
	liquid water	~ 80

substituting

$$\rightarrow \nabla \cdot (\varepsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f$$

$$\nabla \cdot (\varepsilon_0 \mathbf{E} + \varepsilon_0 \chi_e \mathbf{E}) = \rho_f$$

$$\nabla \cdot [\varepsilon_0 (1 + \chi_e) \mathbf{E}] = \rho_f$$

$\mathbf{D}$

Defining $\varepsilon_r = 1 + \chi_e$ relative permittivity [–]

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E}$$

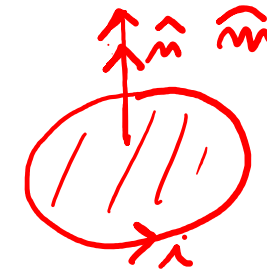
Constitutive
relation

$$\nabla \cdot \bar{\mathbf{E}} = \rho_f$$

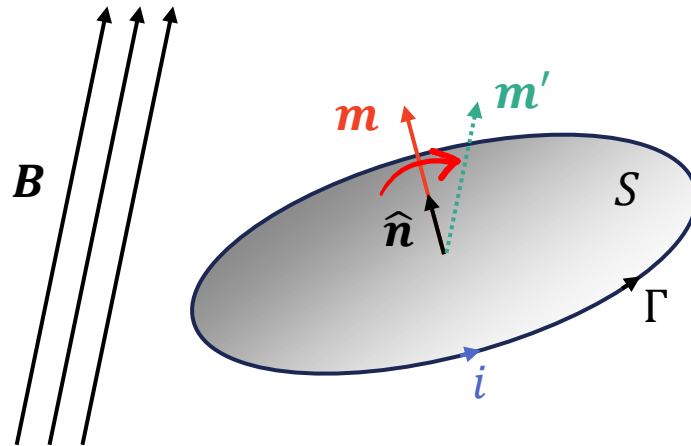
$$\left\{ \begin{array}{l} \nabla \cdot \bar{\mathbf{D}} = \rho_f \\ \bar{\mathbf{D}} = \underbrace{\varepsilon_0 \varepsilon_r}_{\varepsilon} \bar{\mathbf{E}} \end{array} \right.$$

Magnetization – microscopic description

Atom/molecules → electrons in motion (atomic currents)



closed circuit of atomic dimensions → can be modelled as **elementary magnetic dipole**



current SURFACE of elementary current loop

$$\mathbf{m} = i\Delta S\hat{n} \quad \text{magnetic dipole moment}$$

$$[\mathbf{m}] = \text{Am}^2$$

Magnetization – macroscopic description

Goal: obtain **macroscopic description** of magnetization

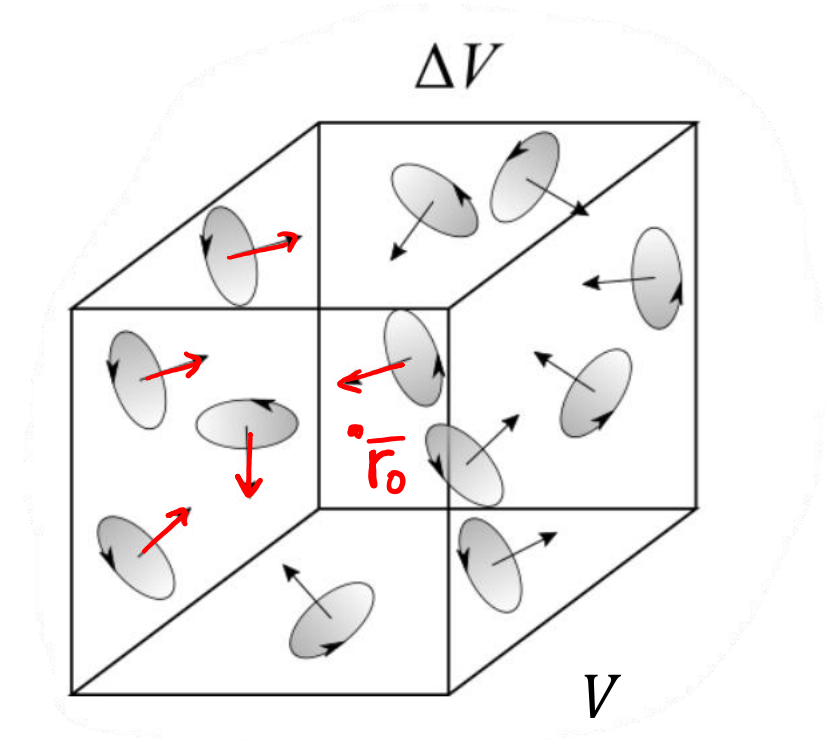
Let $\mathbf{r}_0$ be a point in a region V with volume ΔV

$$\mathbf{M}(\mathbf{r}_0) = \lim_{\Delta V \rightarrow 0} \frac{\Delta \mathbf{m}}{\Delta V} \quad [\mathbf{M}] = \frac{\text{A m}^2}{\text{m}^3} = \frac{\text{A}}{\text{m}}$$

(Note: In the original image, $\Delta \mathbf{m}$ and the final $\frac{\text{A}}{\text{m}}$ term are circled in red, and a blue line connects $\Delta \mathbf{m}$ to the limit expression.)

Magnetization field: magnetic dipole moment per unit volume
measures the local strength and direction of magnetization

Problem: magnetization produces **bound currents** inside matter, and these currents contribute to $\mathbf{B}$



Magnetization field

A distribution of magnetic dipoles can be represented by an equivalent *magnetization current density*

$$\nabla \times \mathbf{M} = \mathbf{J}_m$$

$\uparrow$
 A/m

$\mathbf{J}_m$: magnetization current density [A/m²]

$$\frac{\partial \bar{\mathbf{P}}}{\partial t} = \mathbf{J}_p$$

$\mathbf{J}_p$: polarization current density [A/m²]

...we can fix the Ampère's law!

$$\frac{\text{C}}{\text{m}^2} \frac{1}{\text{s}} = \frac{\text{A}}{\text{m}^2}$$

in free space

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) = \mathbf{J}_f + \frac{\partial \epsilon_0 \mathbf{E}}{\partial t}$$

in free space, all
current is free current

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_m + \mathbf{J}_p$$

→ in (magnetizable) matter

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) = \mathbf{J}_f + \mathbf{J}_m + \frac{\partial \epsilon_0 \mathbf{E}}{\partial t} + \mathbf{J}_p$$